

EXP:6

DATA VISUALIZATION USING POWER BI

AIM:

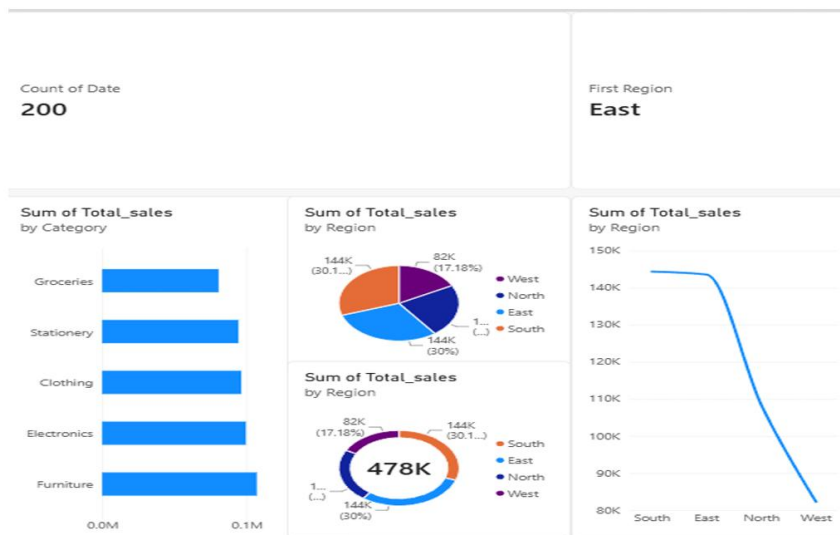
To learn the Power BI Desktop interface and workflow by connecting Excel/CSV data sources, performing basic data transformation, creating calculated columns and DAX measures, and developing interactive visualizations and dashboards.

PROCEDURE:

1. Open **Power BI Desktop** and explore the interface, including the Fields, Visualizations, Filters, Data, Report, and Model views.
2. Connect to the required data sources:
 - Select **Home → Get Data → Excel** and import SalesData.xlsx.
 - Select **Home → Get Data → Text/CSV** and import Targets.csv.
 - Select the required tables or sheets and click **Load**.
3. Open **Transform Data** to launch Power Query Editor.
4. Clean and transform the data:
 - Set the **Date** column to the Date data type.
 - Create a **Total Sales** calculated column using:
Quantity × UnitPrice
 - Trim unnecessary spaces from text columns.
 - Remove duplicate records if required.
 - Ensure the Month field in Targets.csv has a consistent format.
5. Click **Close & Apply** to load the transformed data into Power BI.
6. Create basic visualizations:
 - **Bar Chart:** Display Sales by Category.
 - **Line Chart:** Display Monthly Sales Trend.
 - **Pie/Donut Chart:** Display Sales proportion by Region.
 - **Card Visuals:** Display Total Sales, Total Quantity, and Average Order Value.
 - **Table/Matrix:** Display the Top 10 Products based on Sales.

7. Add a **Region or Date slicer** to make the report interactive.
8. Arrange the visualizations neatly on a report page to create a dashboard-like layout.
9. Add suitable titles, data labels, tooltips, number formats, and other formatting options.
10. Save the Power BI report as a .pbix file.

OUTPUT:



RESULT:

Thus, the Sales and Target data were successfully connected, transformed, visualized, and presented as an interactive dashboard using Power BI.